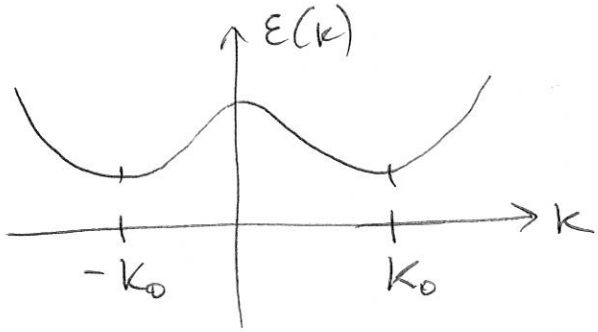
Consider BEC in case of two degenerate minima



$$\varepsilon(\vec{k}_0) = \varepsilon(-\vec{k}_0)$$

$\Rightarrow$ can have condensate ψ_1: $a_k = \sqrt{N}\,\psi_1\,\delta_{\vec{k},\vec{k}_0} + \tilde{a}_k$

or ψ_2: $a_k = \sqrt{N}\,\psi_2\,\delta_{\vec{k},-\vec{k}_0} + \tilde{a}_k$

$$\Rightarrow \quad a_k = \sqrt{N}\,\psi_1\,\delta_{\vec{k},\vec{k}_0} + \sqrt{N}\,\psi_2\,\delta_{\vec{k},-\vec{k}_0} + \tilde{a}_k$$

Then expect

$$\frac{E}{N} = -\mu\left(|\psi_1|^2 + |\psi_2|^2\right) + \frac{1}{2}u\left(|\psi_1|^4 + |\psi_2|^4\right) + v\,|\psi_1|^2\,|\psi_2|^2$$

$$\rho_1 = |\psi_1|^2, \quad \rho_2 = |\psi_2|^2$$

$$\frac{E}{N} = -\mu\left(\rho_1 + \rho_2\right) + \frac{1}{2}u\left(\rho_1^2 + \rho_2^2\right) + v\,\rho_1\rho_2$$

① $\frac{\partial E}{\partial \rho_1} = -\mu + u\rho_1 + v\rho_2 = 0$

$\frac{\partial E}{\partial \rho_2} = -\mu + u\rho_2 + v\rho_1 = 0$

$$\left.\begin{aligned} u\rho_1 + v\rho_2 &= \mu \\ u\rho_2 + v\rho_1 &= \mu \end{aligned}\right\} \qquad \rho_1 = \frac{\mu - v\rho_2}{u}$$

$$u\rho_2 + \frac{v}{u}(\mu - v\rho_2) = \mu$$

$$(u^2 - v^2)\rho_2 = u\mu - v\mu$$

$$(u-v)(u+v)\rho_2 = (u-v)\mu$$

$$\Rightarrow \quad \rho_2 = \frac{\mu}{u+v}$$

$$\rho_1 = \frac{1}{u}\left(\mu - \frac{v\mu}{u+v}\right) = \frac{1}{u}\,\frac{u\mu}{u+v} = \frac{\mu}{u+v}$$

$$\Rightarrow \quad \frac{E}{N} = -\mu\,\frac{2\mu}{u+v} + \frac{1}{2}u \cdot 2\left(\frac{\mu}{u+v}\right)^2 + v\left(\frac{\mu}{u+v}\right)^2$$

$$= -\frac{2\mu^2}{u+v} + (u+v)\frac{\mu^2}{(u+v)^2} = -\frac{\mu^2}{u+v}$$

② Another solution: $\rho_2 = 0 \Rightarrow \rho_1 = \frac{\mu}{u}$

$$\frac{E}{N} = -\mu \cdot \frac{\mu}{u} + \frac{1}{2}u\,\frac{\mu^2}{u^2} = -\frac{\mu^2}{2u}$$

$$-\frac{\mu^2}{2u} < -\frac{\mu^2}{u+v}$$

$$\frac{1}{2u} > \frac{1}{u+v}$$

$$u+v > 2u \Rightarrow v > u$$

$\Rightarrow$ if $\underline{v > u}$ we have usual single-condensate solution $\begin{cases} \rho_1 = \frac{\mu}{u}, \ \rho_2 = 0 \\ \text{or } \rho_1 = 0, \ \rho_2 = \frac{\mu}{u} \end{cases}$

if $\underline{u > v}$ $\quad \rho_1 = \rho_2 = \frac{\mu}{u+v}$

double-condensate!

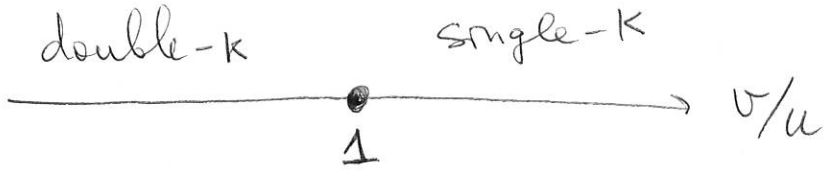
if $\underline{u = v}$: $\frac{E}{N} = -\mu(\rho_1+\rho_2) + \frac{1}{2}u(\rho_1^2+\rho_2^2+2\rho_1\rho_2)$

$$= -\mu(\rho_1+\rho_2) + \frac{1}{2}u(\rho_1+\rho_2)^2$$

point of extended symmetry

$$\rho_1+\rho_2 = \rho = \frac{\mu}{u}$$

but ρ_1 and ρ_2 can be distributed arbitrary.



transition point

$$\psi_1 = \sqrt{\rho_1}\, e^{i\theta_1}$$

$$\psi_2 = \sqrt{\rho_2}\, e^{i\theta_2}$$

$$S_r^+ = 2S\sqrt{\frac{2\mu}{u+v}}\, e^{i\theta_+} \cdot \cos\left[\vec{K}_0 \cdot \vec{r} + \theta_-\right]$$

$$S_r^z = S - \frac{8\mu}{u+v} \cos^2\left[\vec{K}_0 \cdot \vec{r} + \theta_-\right]$$

$$\theta_{1,2} = \theta_+ \pm \theta_-$$

$$\theta_+ = \frac{1}{2}(\theta_1 + \theta_2)$$

$$\theta_- = \frac{1}{2}(\theta_1 - \theta_2)$$

umbrella spin structure

$$S_r^+ = S\sqrt{\frac{2\mu}{u}}\, e^{i\vec{K}_0 \cdot \vec{r} + i\theta_1}$$

$$S_r^z = S - \frac{\mu}{u}$$

$$\psi_1 = \sqrt{\rho_1}\, e^{i\theta_1}$$